7.2 Transverse Waves: EM Spectrum & Polarisation

Question Paper

Course	CIE A Level Physics
Section	7. Waves
Topic	7.2 Transverse Waves: EM Spectrum & Polarisation
Difficulty	Medium

Time allowed: 20

Score: /10

Percentage: /100

Question 1

Two lasers are set up to pass through a vacuum. One laser emits red light; the other emits green light.

Which property of the two lasers are different?

A. speed

B. plane of polarisation

C. frequency

D. amplitude

[1 mark]

Question 2

The wavelength of light emitted by a laser is 600 nm.

Which of the following is the distance in number of wavelengths, that is travelled by light in one second?

A. $5.0 \times 10^{14}\ s^{-1}$

B. $5.0 \times 10^{8}\ s^{-1}$

C. $5.0 \times 10^{11}\ s^{-1}$

D. $5.0 \times 10^{17}\ s^{-1}$

[1 mark]

Question 3

Waves of the electromagnetic spectrum travel through a vacuum at speed c and with a wavelength λ and frequency f

Which of the following describes the speed and wavelength when the frequency is $\frac{f}{2}$?

	λ	c
A	2λ	$2c$
B	2λ	c
C	$\frac{\lambda}{2}$	c
D	$\frac{\lambda}{2}$	$\frac{c}{2}$

[1 mark]

Question 4

The diameter of a nucleus and the wavelength of an electromagnetic wave are in the same order of magnitude.

In which region of the electromagnetic spectrum would that wavelength occur?

A. infrared

B. visible light

C. x-ray

D. gamma-ray

[1 mark]

Question 5

R is a source emitting infra-red radiation and S is a source emitting ultra-violet radiation. The table suggests the wavelengths for R and S.

Which row is correct?

	Wavelength emitted by R / m	Wavelength emitted by S / m
A	5×10^{-7}	5×10^{-8}
B	5×10^{-8}	5×10^{-10}
C	5×10^{-5}	5×10^{-8}
D	5×10^{-5}	5×10^{-10}

[1 mark]

Question 6

Which statement would be deduced from a table of wavelengths of the waves in the electromagnetic spectrum?

A. the wavelength range for radio waves is less than that for infrared waves

B. the wavelength range for x-rays is less than that for radio waves

C. red light has a shorter wavelength than green light

D. green light has a shorter wavelength than x-rays

[1 mark]

Question 7

A telescope detects electromagnetic radiation with wavelengths 2 cm.

Which type of telescope is it ?

A. x-ray telescope

B. optical telescope

C. microwave telescope

D. radio telescope

[1 mark]

Question 8

A linearly polarised electromagnetic beam has an intensity of 16 W m^{-2} with an angle of incidence of 45° to the vertical.

What is the intensity of the transmitted beam?

A. 20 W m^{-2}

B. 12 W m^{-2}

C. 16 W m^{-2}

D. 8 W m^{-2}

[1 mark]

Question 9

Unpolarised light of intensity I_0 is incident on a polariser with a vertical transmission axis. The transmitted light is incident on a sheet of material X. After transmission through X the intensity of light is $\frac{I_0}{2}$

It is suggested that X could be

1 a polariser with a vertical transmission axis

2 a polariser with a horizontal transmission axis

3 non-polarising glass

A. 1 & 2

B. 2 & 3

C. 1 & 3

D. 3 only

[1 mark]

Question 10

The quality of the sound produced by a portable radio can be improved by changing the orientation of the aerial.

Which statement will correctly describe the explanation of this improvement?

A. the radio waves become polarised as a result of adjusting the aerial

B. the radio waves become unpolarised as a result of adjusting the aerial

C. the radio waves from the transmitter are unpolarised

D. the radio waves from the transmitter are polarised

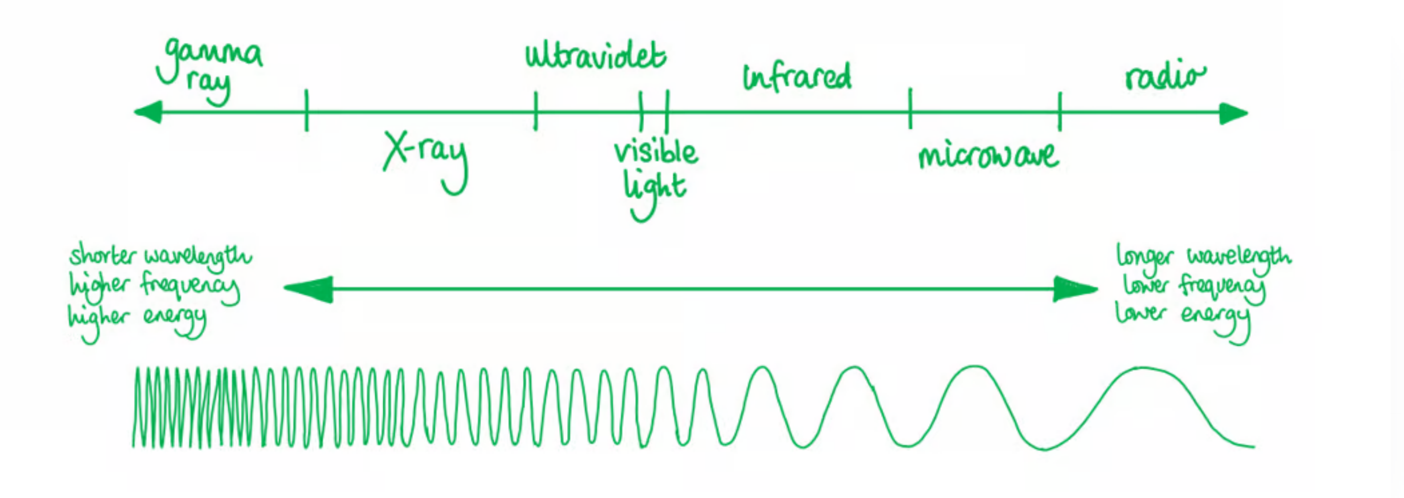
[1 mark]

Model Answers: Medium

1

The correct answer is **C** because:

- Visible light is part of the electromagnetic spectrum



- A laser is a highly directional beam of light
- The **frequency** of the light determines the colour of the laser light
- Since red and green are on different parts of the spectrum, they have different frequencies

A is incorrect because the speed of all EM waves in a vacuum is the same (3.0×10^8 m s^{-1})

B is incorrect because laser light is not polarised light but light directed into a narrow beam of light, polarised light is light that is travelling in one plane only

D is incorrect because the amplitude of the light wave determines the intensity or brightness of the wave, it tells you how much energy the wave carries

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The correct answer is **A** because:

- Wavelength is defined as the distance between successive crests of a wave
- Frequency of a wave is defined as the number of waves passing a point in one second (the same as the definition given in the question)
- The wave equation that links speed, wavelength and frequency is:
  - $v = \lambda \times f$
- The speed of light is 3.0×10^8 m s^{-1}
- The wavelength is 6.0×10^{-7} m
  - Therefore, the frequency is equal to $= \frac{3.0 \times 10^8}{6.0 \times 10^{-7}} = 5.0 \times 10^{14}\ Hz$

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The correct answer is **B** because:

- Wavelength is defined as the distance between successive crests of a wave
- Frequency of a wave is defined as the number of waves passing a point in one second
- The wave equation that links speed, wavelength and frequency is:
  - $v = \lambda \times f$
- The speed *c* of all EM waves in a vacuum is constant
- So, a change in *f* will only cause *λ* to change
  - $c = f\lambda$
  - $c = \left(\frac{f}{2}\right)(2\lambda)$
- Therefore, if the frequency of the wave is halved, then the wavelength will double

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The correct answer is **D** because:

- The diameter of a hydrogen nucleus is 1.76×10^{-15}m
- The wavelength of a gamma-ray is in the range of 10^{-10} to 10^{-16} m
- Therefore, gamma rays have a wavelength similar to the diameter of a hydrogen nucleus

A is incorrect because **infrared** waves have a wavelength in the range: 10^{-3} to 7×10^{-7} m

B is incorrect because **visible light** has waves with a wavelength in the range: 4×10^{-7}

to 7×10^{-7} m

C is incorrect because **x-rays** have a wavelength in the range: 10^{-8} to 4×10^{-13} m

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The correct answer is **C** because:

- Source R is emitting infra-red radiation; infra-red radiation has a wavelength in the range of 10^{-3} to 7×10^{-7} m
- Source S is emitting ultra-violet radiation, ultra-violet radiation has a wavelength in the range of 4×10^{-7} to 10^{-8} m
- The only answer that would fit both of these ranges is option **C**

A is incorrect because

R is in the wavelength range of visible light

S is in the wavelength range of ultra-violet radiation

B is incorrect because

R is in the range of ultra-violet radiation

S in the range of the wavelengths of x-rays

D is incorrect because R is in the correct range, but S is in the range of the wavelengths of x-rays

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The correct answer is **B** because:

- Wavelength is defined as the distance between successive crests of a wave
  - The wavelength of x-rays is in the range $1 \times 10^{-8} - 1 \times 10^{-13}$ m
  - The wavelength of radio waves is greater than 1×10^{-1} m
- Therefore, the wavelength range for x-rays is less than that for radio waves

A is incorrect because the wavelength of radio waves is greater than 1×10^{-1} m, the wavelengths of infrared waves is in the range of $1 \times 10^{-3} - 7 \times 10^{-7}$ m

C is incorrect because the wavelength of red light is 620 – 750 nm whereas green light is shorter with a range of 495 – 570 nm

D is incorrect because the wavelength of green light is in the range of 495 – 570 nm, whereas x-rays are in the range of $1 \times 10^{-8} - 4 \times 10^{-13}$ m

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The correct answer is **C** because:

- Wavelength is defined as the distance between successive crests of a wave
  - Wavelengths of 2 cm are 0.02 m or 2×10^{-2} m
  - The wavelengths of microwaves are in the range of $0.1 - 1 \times 10^{-3}$ m
- Therefore, the telescope detects microwave radiation

A is incorrect because x-rays are in the range: $1 \times 10^{-8} - 4 \times 10^{-13}$ m

B is incorrect because visible light rays are in the range: $4 \times 10^{-7} - 7 \times 10^{-7}$ m

D is incorrect because radio waves have wavelengths greater than 1×10^{-1} m

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The correct answer is **D** because:

- Malus's law allows the calculation of the intensity of the beam of plane polarised light, it is given by
  - $I = I_0 \cos^2 \theta$
- The initial intensity is equal to $I_O = 16$ W m^{-2}
- The angle of incidence is $\theta = 45°$
  - $I = 16 \times \cos^2 45 = 8\ W\ m^{-2}$
- Therefore, the intensity of the transmitted beam is 8 W m^{-2}

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The correct answer is **C** because:

- Polarisation is the process by which oscillations are made to occur in one plane only
- The half rule states that when light passes through the first polarising filter it will

9

The correct answer is **C** because:

- Polarisation is the process by which oscillations are made to occur in one plane only
- The half rule states that when light passes through the first polarising filter it will reduce the intensity by $\frac{1}{2}$ so the intensity drop of $\frac{I_0}{2}$ will be caused by the first polariser
- If the first polarising sheet is on a vertical plane, then material X could be either another vertical plane or a non-polarising glass, because neither option will change the polarisation or affect the intensity of the transmitted light

A & B are incorrect because if X was a polariser with a horizontal transmission axis, then the intensity of the transmitted light would be zero

D is incorrect because since the light will be polarised vertically after passing through the first filter, options 1 and 3 are both valid

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The correct answer is **D** because:

- Polarisation is the process by which the oscillations are made to occur in one plane only
- Radio waves emitted by an antenna are usually transmitted with a specific polarisation, and therefore, receiving aerials will generally only receive radio waves of this polarisation
- If the aerial is moved and this improves the signal then this proves the radio waves must be polarised

A is incorrect because the waves will not become polarised as a result of adjusting the aerial

B is incorrect because the waves will not become unpolarised as a result of adjusting the aerial

C is incorrect because if the waves from the transmitter where unpolarised then an adjustment on the aerial would not make a difference as waves would be travelling in all directions